

UrbanPulse

Smart-City Analysis Report

RDBMS-backed data quality, descriptive analysis, model review, and provenance assessment

Final classification
Fit only for descriptive reporting
Fit only for descriptive reporting

Source workbook: 02_SmartCities_UrbanPulse_RAW-2.xlsx

Observed period: 2026-01-01 to 2026-09-24

Prepared from the completed UrbanPulse Steps 1-9 workflow

Executive summary

Decision

The dataset is **fit only for descriptive reporting**. It is appropriate for aggregate summaries, exploratory dashboards, data-quality demonstrations, training, and pipeline testing. It should not be used alone for operational control, safety decisions, causal claims, KPI commitments, or production forecasting.

Verification boundary

This report includes only findings verified from the supplied raw workbook, the SQLite RDBMS, and the completed analysis workflow. The original project PDF was unavailable, so this report deliberately does not state the original project objective, official field definitions, original sampling narrative, external source documentation, original charts, original conclusions, or any claims that could only be verified from that document.

Raw rows	Canonical rows	Eligible rows	Provenance flags
1,635	1,600	1,491	0

Main findings

- The raw layer contains 1,635 rows. The canonical analytical layer contains 1,600 rows after exact-duplicate handling, with 35 duplicate rows retained in the audit trail.
- The sample has one canonical row per unique timestamp after deduplication. It is intentionally sampled and is not a complete city-zone-time panel.
- City coverage is relatively balanced, while zone coverage is uneven. Central and North contain most labeled observations, and 32 rows have unknown zones.
- None of the eight registered hypotheses was supported at the predefined 99% threshold. The fitted relationships also had 99% intervals that included the null effect.
- Forecasting models did not provide reliable improvement over simple chronological baselines. Provenance testing produced zero flags under the independent-permutation null, which is not proof of authenticity.

1. Data architecture and analytical design

Two-layer data model

The workflow preserves an immutable raw layer and a canonical cleaned layer. The canonical layer keeps raw and cleaned values side by side, records quality flags, and logs changes separately. Numeric values were not silently corrected or imputed in the primary analysis.

Layer	Purpose	Rows	Treatment
raw_urbanpulse	Immutable source record	1,635	Values preserved exactly
canonical_cleaned	Primary analytical base	1,600	Exact duplicates handled; values and flags retained
imputed sensitivity layer	Sensitivity-only comparison	1,600	135 imputation events; never used as primary result
quality_flag and change_log	Auditability	n.a.	Missing, imputed, corrected, outlier, and unknown states recorded separately

Primary analysis policy

The primary statistical analysis uses observed values only. Missing values are not imputed. Mean, median, and mode imputation are retained as a separate sensitivity analysis so that their effect can be compared without altering the primary evidence.

Quality decision rule

Confirmed errors are corrected only when supported. Valid extremes are retained and flagged. Unknown values are preserved and excluded or analyzed separately. The speed value -10 was not changed to +10, and questionable temperatures were not replaced by generic internet averages.

Scope and uncertainty

The analysis uses a 99% confidence standard, meaning $\alpha = 0.01$ for the predefined tests. This is a statistical threshold, not a guarantee that the conclusions are 99% factually certain.

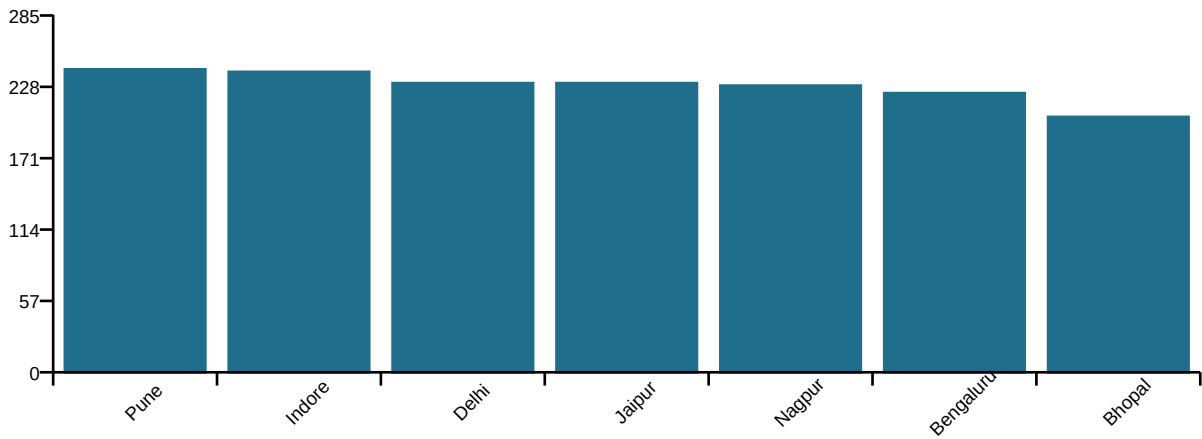
2. Coverage and pivot analysis

Pivot table: observations by city and zone

This table describes the distribution of the sampled rows. It should not be interpreted as a complete city-zone panel.

City	(missing)	Central	East	North	South	West	Total
Pune	4	68	35	62	31	42	242
Indore	1	66	35	66	34	38	240
Delhi	8	58	39	69	25	32	231
Jaipur	5	62	33	67	25	39	231
Nagpur	5	74	34	64	24	28	229
Bengaluru	4	61	34	50	38	36	223
Bhopal	5	60	27	59	30	23	204
Total	32	449	237	437	207	238	1,600

Canonical observations by city



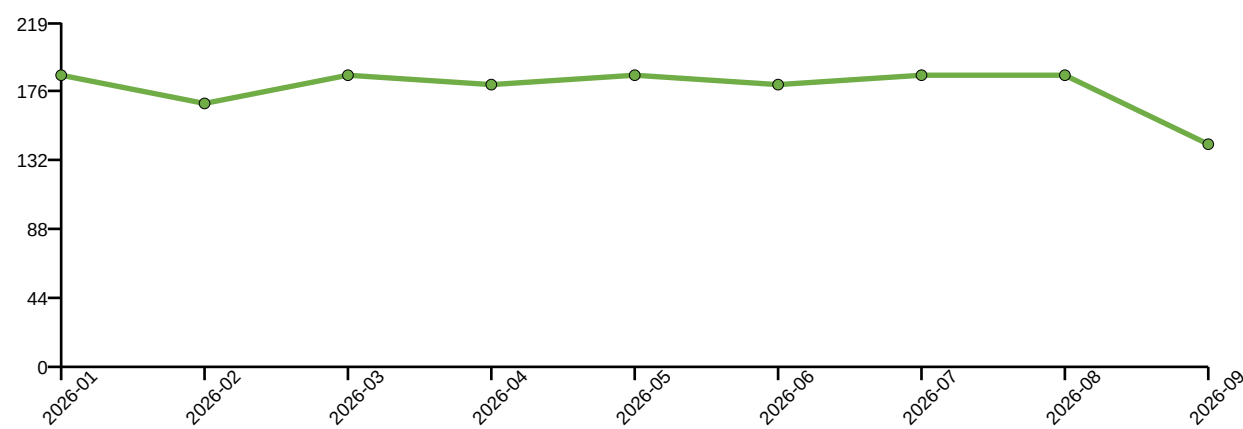
3. Time coverage and metric distributions

Pivot table: observations by month and four-hour slot

The timestamp grid is balanced across six four-hour slots. September is partial because the source ends on 2026-09-24 at 12:00.

Month	00:00	04:00	08:00	12:00	16:00	20:00	Total
2026-01	31	31	31	31	31	31	186
2026-02	28	28	28	28	28	28	168
2026-03	31	31	31	31	31	31	186
2026-04	30	30	30	30	30	30	180
2026-05	31	31	31	31	31	31	186
2026-06	30	30	30	30	30	30	180
2026-07	31	31	31	31	31	31	186
2026-08	31	31	31	31	31	31	186
2026-09	24	24	24	24	23	23	142
Total	267	267	267	267	266	266	1,600

Canonical observations by month



The chart shows observation counts, not an outcome trend. Changes in counts can reflect the intentional sampling design or the partial final month.

4. Descriptive metric results

Pivot table: descriptive metric summary

The values below use the canonical observed layer. Invalid speed values are excluded from speed-specific summaries according to the data contract. Other flagged extremes remain visible and documented.

Metric	n	Mean	Median	SD	Min	Max	99% CI
Average speed (km/h)	1,551	38.2	37.9	11.8	3.0	85.1	[37.4, 39.0]
Citizen complaints	1,600	8.0	8.0	2.8	1.0	17.0	[7.8, 8.1]
Parking occupancy (%)	1,600	68.1	67.9	20.3	5.0	110.0	[66.9, 69.4]
Power outage (minutes)	1,600	11.9	8.0	12.4	0.0	127.2	[11.1, 12.7]
Public transport usage	1,600	1,298.7	1,305.0	694.0	100.0	2,498.0	[1,253.5, 1,343.8]
Road incidents	1,600	2.0	2.0	1.4	0.0	8.0	[1.9, 2.1]
Temperature (C)	1,568	28.9	28.9	7.0	6.4	53.8	[28.5, 29.4]
Traffic volume	1,568	2,050.9	2,008.5	1,100.1	214.0	3,999.0	[1,980.2, 2,122.6]
Waterlogging reports	1,600	1.0	1.0	1.0	0.0	6.0	[1.0, 1.1]

Pivot table: mean metrics by city

City	Traffic	Speed	Parking %	Temp C	Complaints
Bengaluru	2,110.3	38.8	67.4	28.8	8.0
Bhopal	2,184.2	37.7	67.5	28.9	7.9
Delhi	1,926.7	38.1	67.4	29.4	8.0
Indore	2,015.0	38.1	68.7	28.7	7.6
Jaipur	2,105.5	38.2	68.2	28.7	8.1
Nagpur	2,073.5	37.7	69.5	29.3	8.1
Pune	1,966.9	38.5	68.1	28.7	7.9

City means are descriptive summaries of an intentionally sampled stream. They are not official city rankings and should not be interpreted as synchronized comparisons.

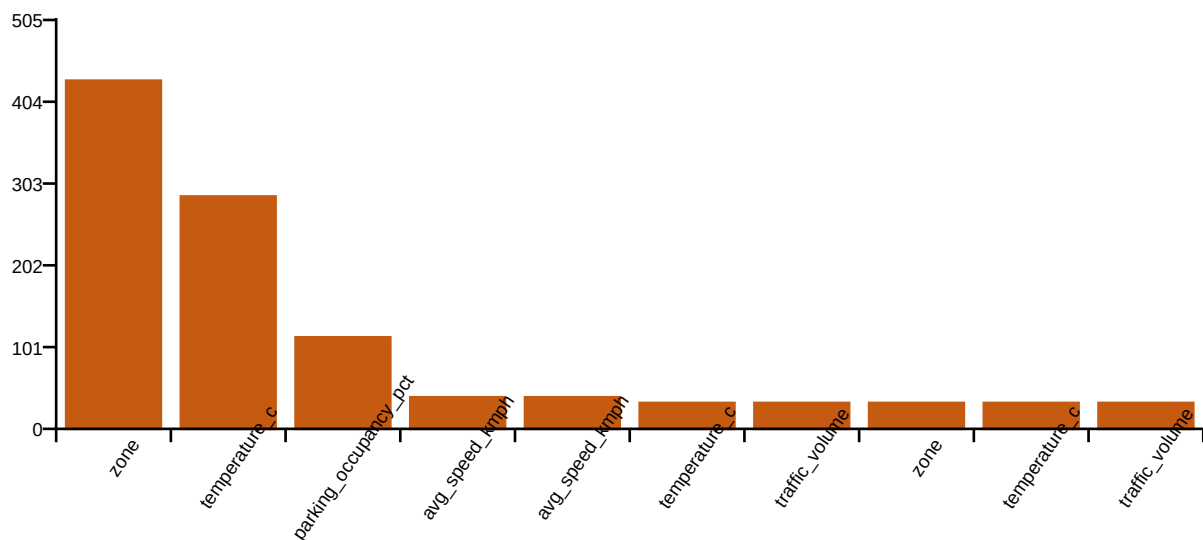
5. Data quality and validation

Recorded quality signals

The database keeps raw values intact and records the reason for each flag. The largest quality signal is zone casing normalization, which is a canonicalization rather than a numeric correction. Temperature and parking flags require domain context before operational use.

Flag type	Field	Count
corrected	zone	430
outlier	temperature_c	287
outlier	parking_occupancy_pct	113
imputed	avg_speed_kmph	39
missing	avg_speed_kmph	39
imputed	temperature_c	32
imputed	traffic_volume	32
imputed	zone	32
missing	temperature_c	32
missing	traffic_volume	32
missing	zone	32
unknown	zone	32
outlier	power_outage_minutes	16
outlier	avg_speed_kmph	10

Largest recorded quality-flag counts



Temperature validation used month plausibility, neighboring observations from the same city, and benchmark checks. No generic internet average was used as a replacement. Remaining uncertain values are preserved and flagged.

6. Hypothesis tests and model fitting

Predefined hypothesis tests

The tests used the predefined 99% threshold and multiple-testing control where applicable. None of the eight registered hypotheses was supported. These are non-detection results, not proof that all variables are mathematically independent.

ID	Relationship	Spearman	99% interval	Holm p	Supported
H1	traffic_volume vs avg_speed_kmph	0.014	[-0.052, 0.080]	1.000	No
H2	traffic_volume vs road_incidents	-0.016	[-0.081, 0.049]	1.000	No
H3	waterlogging_reports vs avg_speed_kmph	-0.010	[-0.076, 0.055]	1.000	No
H4	waterlogging_reports vs road_incidents	0.018	[-0.047, 0.082]	1.000	No
H5	temperature_c vs power_outage_minutes	0.001	[-0.064, 0.066]	1.000	No
H6	temperature_c vs citizen_complaints	0.005	[-0.061, 0.070]	1.000	No
H7	power_outage_minutes vs citizen_complaints	-0.032	[-0.097, 0.032]	1.000	No
H8	road_incidents vs avg_speed_kmph	-0.013	[-0.078, 0.053]	1.000	No

Adjusted relationships

Target	Predictor	Model	Effect	99% interval
avg_speed_kmph	traffic_volume	huber	0.1978	[-0.4043, 0.7986]
road_incidents	waterlogging_reports	poisson	IRR 1.016	[0.967, 1.064]
power_outage_minutes	temperature_c	huber	-0.0189	[-0.1230, 0.0868]

Forecasting

Rolling chronological validation did not establish useful generalization. The mean R-squared across forecast metric folds was -1.508. Some individual metrics showed small MAE improvements, but intervals included no improvement and several targets performed materially worse than their simple baseline.

7. Provenance check

Simulation design

The canonical observed layer was compared with 5,000 simulated datasets. Each simulation independently permuted observed non-missing metric values while preserving the row count, marginal distributions, missingness mask, timestamp grid, city labels, and zone labels. This breaks cross-field, group, and temporal alignment while holding one-dimensional distributions fixed.

Diagnostic family	Purpose	Result
Pairwise absolute Spearman	Detect unusual cross-field alignment	No flags
City and zone eta-squared	Detect unusual group alignment	No flags
Within-city lag-1 autocorrelation	Detect unusual temporal alignment	No flags
Entropy and unique-rate	Confirm marginal preservation	Fixed by construction
Raw duplicate structure	Report source-layer duplication separately	35 duplicate rows; canonical layer deduplicated

Interpretation boundary

Zero flags mean the observed relationships were not unusual relative to this independent-permutation null model. This does not prove that the data are authentic, operationally collected, or non-synthetic. Source documentation, collection metadata, and system provenance remain necessary for a provenance conclusion.

8. Final decision and recommended use

Final classification: Fit only for descriptive reporting

This classification reflects the intentional sample design, unresolved semantic and source decisions, weak evidence for adjusted relationships, and unreliable forecasting performance. The absence of provenance flags does not upgrade the dataset to operational fitness.

Permitted use	Conditions
Descriptive reporting	Aggregate summaries, ranges, distributions, coverage analysis, and exploratory charts with limitations disclosed.
Training and pipeline testing	Use as a controlled educational or engineering dataset. Keep the intentional sample and quality flags visible.
Operational analytics	Not approved from this dataset alone. Requires representativeness evidence, owner confirmation of metric semantics, source or station metadata, and repeat validation.
Causal or policy claims	Not supported. The data structure and results do not establish causality.
Production forecasting	Not supported. Rolling validation did not show reliable generalization.

Upgrade conditions

Before operational use, obtain the sampling frame and representativeness documentation, confirm the meaning and valid ranges for speed, parking, outage, temperature, and zone, attach source or station metadata, resolve unknown quality decisions, and rerun the validation workflow on the remediated data.

Appendix: reproducibility and deliverables

Reproducibility record

The analysis is recorded in SQLite with raw, canonical, quality, sensitivity, model, provenance, and decision tables. The raw workbook was not overwritten. The report and companion workbook are generated from the same database summaries.

Artifact	Purpose
urbanpulse_rdbms.sqlite	RDBMS source of record for the completed workflow
urbanpulse_analysis_pivots_charts.xlsx	Pivot tables and editable chart views
urbanpulse_analysis_report.pdf	This narrative analysis report
urbanpulse_step5_compact_dashboard.html	Compact descriptive dashboard
urbanpulse_step8_provenance_check.md	Detailed provenance simulation result
urbanpulse_step9_fitness_decision.md	Recorded final fitness decision

SQLite integrity check: ok

Raw data and canonical values remain unchanged by report generation.